EL NIÑO ACTION BRIEFING · 26 SEPTEMBER 2026 · VERIFIED RUN · COFFEE-INDUSTRY ACTIONS FIRST

Jinotega, Nicaragua

In Jinotega's wet highlands, El Niño arrives as a lower cup score, more borer, and pickers gone to the gold mines, not as an empty tree. Its drier west is the drought zone. Last time, the harvest's collapse came from a bankrupt exporter, not the weather. Every crisis here has been survived by whoever held credit.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Jinotega department, about 480,000–500,000 people (1–2 Climate Impact Lab regions)	No figure published for Nicaragua	Jun 2026 – Feb 2027 (heat only)	West: drought. Highlands: heat, borer, quality loss. East: landslide, malaria. All: pickers leaving

13/13 recommendations verified. 6 corrected (partially confirmed), 0 confirmed outright, 3 unverified, 4 contraindicated. **The funding target moved by more than a third**, from $362k to $131k (down 64%): only three Jinotega estates with picker payrolls could be named, and the highland drought case for nutrition spending did not hold. **Two lens claims about rainfall were struck**: an April report of severe early deficits came from someone working in the south, and a cited August weather-service statement could not be found. **Ranking rule:** coffee-industry actions first, at the reader's request, all filtered for political safety. **Resolution caveat:** a department, not a town; the actions name municipalities. **Method caveat:** web search was exhausted; agents worked by direct fetch.

EXECUTIVE SUMMARY — AT A GLANCE



13.09°N 86.00°W, Nicaragua · Natural Earth

- **The risk.** Drought in the west and heat, borer, and unfilled beans in the highlands; the 2023 El Niño cut output 10% and exports 15%, and Iota killed three in Wiwilí in 2020.
- **First action.** Importers and roasters split 2026/27 contracts across at least two exporters, or buy direct from cooperatives, under letters of credit or escrow, by 31 October.
- **The money.** $19.8k floor, $131k target, $362k ceiling; harvest-camp water and oral rehydration through estates leads the placements, $9k at floor and $30k at target.
- **Do not.** Do not send money labelled as a donation or run a branded programme: the foreign-agents law reportedly counts foreign company funding, and branded top-ups expose the people they reach.

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* Suggested formulation: check with a health worker before sharing.

National facts carry over from the verified Ocotal run: no drought trigger pays in Nicaragua, more than 5,500 organisations are closed, and every payment must be tied to invoiced coffee.

EXPOSURE, HONESTLY

A DEPARTMENT WITH THREE DIFFERENT WINTERS

- **Where.** Eight municipalities: Jinotega, San Rafael del Norte, San Sebastián de Yalí, La Concordia, Santa María de Pantasma, El Cuá, San José de Bocay, and Wiwilí. Jinotega is one of the three departments, with Matagalpa and Nueva Segovia, that hold more than 85% of Nicaragua's arabica farms (USDA). The widely quoted "65%" or "80%" of national coffee has no source and is not used. About 90% of producers are small or medium.
- **This season's rain is split.** The only dated evidence (Confidencial, 25 September) puts the severe deficit in about 63 Dry Corridor municipalities, with August the worst month and the shortfall expected through October; humid northern zones had near-normal or above-normal rain. Jinotega's drier west (San Rafael del Norte, Yalí, La Concordia) is plausibly in deficit; no official list confirms it. The government's director of meteorology called the deficits present but "not severe," while Honduras, El Salvador, and Panama declared alerts.
- **In the highlands the hazard is quality.** Farmers report heat spikes (34 °C at Peñas Blancas), coffee berry borer, and beans that do not fill and break in the pulper. USDA found the 2023 El Niño cut Nicaragua's 2023/24 output 10% and exports 15%, through both yield and quality.
- **The labour squeeze.** Last harvest, producers raised pay to C$100 or more per lata plus meals, while artisanal gold mining paid up to C$1,200 a day. El Cuá was about 40% short of pickers by late December; estates trucked workers from León, Chinandega, Managua, and La Dalia. The harvest pauses over Christmas. USDA: labour shortages doubled harvest costs in 2024/25.
- **The last collapse was financial.** Mercon, owner of the exporter CISA, filed for bankruptcy on 6 December 2023. CISA stopped receiving coffee at its northern mill days before harvest; it handled about half of Nicaragua's exportable coffee. In January 2024 a court put its assets under a state administrator over a disputed $30 million tax claim, and no reopening has been found. USDA: the uncertainty "led to lower prices for farmers."
- **Health.** In remote communities of Jinotega, León, and Matagalpa, infant mortality was 52–59% lower where trained community volunteers treated pneumonia and diarrhoea (observational, 243 infant deaths in total). Nicaragua's malaria incidence rose from 0.27 per 1,000 at risk in 2014 to 5.62 in 2020, then fell to 1.04 in 2024; transmission concentrates on the North Caribbean coast, which Jinotega's eastern municipalities border. No Jinotega malaria figure exists.

* Suggested formulation: check with a health worker before sharing.

The first seven actions have a coffee-industry owner or payer. Ranked by lives and livelihoods protected per unit of effort, and by how safely money moves. The Historian's finding frames them: in the 2012–13 rust, the worst-hit farmers knew the controls and could not pay for them.

Importers: split 2026/27 contracts across at least two exporters, or buy direct from cooperatives, under letters of credit or escrow

PARTIALLY CONFIRMED | SKEPTIC · ECONOMIST | ADAPTATION INVENTORY: PRECEDENT, US CONTINGENT LOANS KEPT FIRMS OPEN

OWNER	BY WHEN	COST	PREVENTS
Importers and roasters buying Jinotega coffee	Before contracts close, by 31 October	1–2% of invoice for letters of credit (estimate)	Growers losing buyer and credit at once, as in December 2023

When CISA closed, growers were left holding pledged coffee with no buyer, just before harvest. Ask any exporter how leveraged it is before advancing money through it. The Economist's warning completes this: at record prices, a grower who sold forward and comes up short must buy coffee at spot or default, the same squeeze that sank Mercon. Rewriting contract terms before November is the cheapest insurance available.

Estates and buyers: pay the harvest in full and early, before the Christmas pause, with a picker-retention premium passed through estate payroll

PARTIALLY CONFIRMED | FRONTLINE · ECONOMIST · EMERGENCY MGR · PHILANTHROPIST

ADAPTATION INVENTORY: PRECEDENT, NICARAGUA CASH TRANSFER PLUS PRODUCTIVE GRANT

OWNER	BY WHEN	COST	PREVENTS
Estates with picker payrolls (three Mierisch farms confirmed); importers via contract price	Commit by 15 November; pay from first picking, which in high Jinotega runs into late December	$0.05/lb premium (~$6.60 a bag), or $1.50 per picker-day	Pickers leaving for the mines, and cash running out over Christmas

* Suggested formulation: check with a health worker before sharing.

The migrant pickers never meet a cooperative, only the estate foreman and cook, so estate payroll is the only channel that reaches them. $1.50 a day is about 7–14% of a harvest day's pay, assuming 4–8 latas at C$100 (the latas per day is an assumption). Paying before the Christmas pause targets the week when cash runs out and workers leave. Keep it as a price differential on invoiced coffee, not a branded programme.

Lenders: pay pre-harvest advances as fertiliser, fungicide, and borer traps, repaid in coffee, with a deferral clause triggered by the right measure for each zone

PARTIALLY CONFIRMED · HISTORIAN · SKEPTIC · ECONOMIST · EMERGENCY MGR · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	TRIGGER
Exporter-lenders and importers financing estates and cooperatives	In 2026/27 contracts, before 31 October	Carry cost on deferred balances, about 8–12% (estimate)	West: delivered volume >15% short. Highlands: screen, cup, or borer downgrades

Villarreyna et al. (2020) studied 29 plot pairs in Jinotega, El Tuma–La Dalia, and San Ramón: the worst 2012–13 rust fell on farmers with little education, training, advice, or income. "Technical solutions ... are not sufficient"; "the market has a crucial role to play." Paying the advance in inputs tackles that directly. The Skeptic's correction is essential: a yield trigger would miss most of Jinotega, whose highland loss arrives as a downgrade, so the highland trigger must be quality. No published contract with either trigger was found.

Buyers: commit to hold the price tier for highland lots despite heat-related defects this season

UNVERIFIED · NO PRECEDENT FOUND · SKEPTIC · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Specialty importers and roasters	Announce before harvest	The downgrade the buyer would otherwise take	Farmers losing income through the buyer's grading, not the weather

* Suggested formulation: check with a health worker before sharing.

The Skeptic's one thing: in the wet highlands, El Niño's damage arrives as a lower cup score and more borer, and farmers lose money when the buyer downgrades the lot. The most protective commitment is the one nobody proposes, because the buyer pays for it. No precedent in specialty contracts was found.

Estates: make the harvest camp the unit of care: ceramic water filters, oral rehydration salts and zinc, and, in eastern camps, bed nets and a trained volunteer

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Estates, invoiced through the dry mill	Before pickers arrive	$3,000 per estate camp (unit costs unsourced)	Diarrhoea, dengue, and malaria where pickers and children crowd

The Public Health lens's one thing: the harvest brings the people to the risk, moving pickers and children into camps in the malaria-receptive east where volunteer health coverage is thinnest. Ceramic filters cut diarrhoea by 61% (RR 0.39, moderate quality, Cochrane), chlorination by 23%. Insecticide-treated nets cut child deaths 17% (high certainty) and vivax malaria episodes 39% (moderate). No study links El Niño to malaria or dengue in Nicaragua or Honduras. A laminated "empty, scrub, cover" card in the estate kitchen costs almost nothing.*

Estates in Wiwilí, Bocay, El Cuá, and Pantasma: a "do not go back" evacuation rule, picker housing mapped against landslide scars, and stock pre-positioned beyond the washout points

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | ADAPTATION INVENTORY: PRECEDENT, NICARAGUA BRIDGES OFFSET FLOOD INCOME LOSS

OWNER	BY WHEN	COST	PREVENTS
Estate and cooperative managers; mill warehouses	Stock by 30 November; drill before the 2027 rains	$2,000–5,000 per stock site (estimate)	Deaths on the trip home, and isolation when roads go

During Iota in November 2020, three people died in Wiwilí: Carlos and Francisco Carazo, father and son, whose house collapsed after they went back from a shelter for belongings, and a man swept away at El Diamante. In San José de Kilambé 15 of 60 homes were damaged. Bocay, Pantasma, and Wiwilí lost power and drinking water, and opposition mayors said the government left them "alone." The Emergency

* Suggested formulation: check with a health worker before sharing.

Manager's one thing: people died because they went back for their things, not because nobody warned them. Iota and Mitch both struck in La Niña; late 2027 is the window if La Niña follows.

Estates bordering forest: fire breaks and a fire watch by January, paid as farm labour

UNVERIFIED | EMERGENCY MGR | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Estates	31 January	$10–20 per hectare (estimate)	Fire reaching the shade canopy and camps, February to April

No Jinotega fire record was found. Kept because heat spikes are reported and the cost is labour estates already employ.

Fund community health volunteers and medicine stock in the remote east, through the health ministry

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Health ministry; funders via UN agencies	Before harvest	Unpriced	Infant deaths from pneumonia and diarrhoea far from a clinic

* Suggested formulation: check with a health worker before sharing.

The best local evidence in this report: Rivera et al. (2017), in remote "category C" communities of Jinotega, León, and Matagalpa, infant mortality ratio 0.41 (0.21–0.81) where volunteers treated locally. Observational, with wide intervals. The channel is state-controlled; no coffee partner.

Buyers: a quality premium for shade and riverside tree retention in the Lake Apanás catchment

UNVERIFIED HISTORIAN ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Roasters and importers	2027 contracts	Unpriced	Losing the mist, water, and hydropower the catchment supplies

Lake Apanás, flooded around 1960, feeds the Centroamérica plant (two 25 MW turbines); with Santa Bárbara, about 100 MW. The "35% of national supply" figure is historical; hydro is now about 10% of generation. No record of El Niño generation cuts was found. Plausible, untested.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Importers & roasters	Exporter-lenders	Estates (incl. Mierisch)	Cooperatives	Dry mills & warehouses	Nicaraguan legal counsel	Health ministry	UNICEF & UN agencies	Funders	Community health volunteers
1. Split contracts across exporters, under escrow	**A/R**	**C**	**I**	**R**		**C**				
2. Pay harvest early, picker-retention premium	**R**		**A/R**			**C**			**C**	
3. Advances in inputs, zone-triggered deferral	**R**	**A/R**	**I**	**I**		**C**				
4. Hold highland price tier despite defects	**A/R**	**I**	**I**	**I**						
5. Harvest-camp care: water, ORS, nets	**R**		**A/R**		**R**				**C**	**R**
6. Eastern estates: "do not go back" rule			**A/R**	**R**	**R**					
7. Fire breaks and fire watch by January			**A/R**							
8. Health volunteers and medicine in the east							**A**	**R**	**R**	**R**
9. Shade premium in Lake Apanás catchment	**A/R**		**R**	**R**						

What the chart shows. **Estates are accountable for four of the nine rows, and only three Jinotega estates with picker payrolls could be named.** Every picker-facing action (rows 2, 5, 6, and 7) runs through an estate payroll or camp, so naming more estates is the plan's main dependency. **Importers and roasters are accountable for three rows and do the work on three more.** Rows 1, 4, and 9 cost the buyer directly, and row 4 has no precedent in specialty contracts. **Row 8 is the only row with a public owner, and the state controls it.** The health ministry runs the channel and there is no coffee partner, so funders can pay for it but cannot direct it. **Volunteers carry an R on rows 5 and 8.** Row 8 is unpriced, and the eastern camps that need them have the thinnest volunteer coverage.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist's one thing survived verification: money is not the constraint here, the number of payrolls willing to carry it is. Spend the first $5,000 on a Nicaraguan legal opinion and signed contract amendments.

FLOOR
$19.8k

TARGET
$131k

CEILING
$362k

LARGEST UNCERTAINTY
Whether importers can name 7+ more Jinotega estates with picker payrolls

* Suggested formulation: check with a health worker before sharing.

Arithmetic. Floor: the three confirmed Mierisch farms in Jinotega (La Escondida 70 ha, San José 46 ha, Las Delicias 12 ha) × $3,000 camp care = $9,000, plus about 80 pickers (0.63 per hectare, from one Confidencial farm) × 90 days × $1.50 = $10,800; total $19,800. Target: 10 estates × $3,000 = $30,000 plus 750 pickers × 90 × $1.50 = $101,250; total $131,250. Ceiling: the lens's original target, 20 estates and 1,500 pickers plus nutrition screening ($60k) and a cooperative reserve ($39k), $361,500. The target fell 64%, more than a third, because only three estates could be named, the 75-pickers-per-estate assumption was three times too high for those farms, and highland rainfall did not support a malnutrition caseload. Money can be committed by 15 November but mostly disbursed from late December, when high Jinotega ripens.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Mierisch Jinotega estates first, then importer-named estates	Action 5: camp water, oral rehydration, latrines	$9k floor; $30k target	Welfare line on an importer's invoice	Ceramic filters RR 0.39 (Cochrane). Only three estates confirmed; no worker numbers published.	Keep, resized
Same estates' pickers	Action 2: payroll top-up	$10.8k floor; $101k target	Contract-price differential paid through payroll	$1.50 ≈ 7–14% of a harvest day. Red de Protección Social protected spending in the 2001 crisis.	Keep, resized
UNICEF with the health ministry	Malnutrition screening and treatment	$60k	Earmarked contribution	$26–53 per healthy life-year, but no evidence of a highland caseload and no UNICEF Jinotega activity found.	Ceiling only
SOPPEXCCA member households	Fairtrade Premium drought reserve	$39k	Committee vote	Site stale since 2014 and currently down; 650 households unverified; highland, so drought is the wrong hazard.	Ceiling only

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Do not route early payment or credit through a single large exporter

CONTRAINDICATED SKEPTIC · ECONOMIST; CONFIRMED

CISA handled about half of Nicaragua's exportable coffee. It stopped buying days before the 2023 harvest; its group owed about $363 million, $202.5 million to Rabobank; its Nicaraguan assets went under state administration; no buyer or reopening has been found. Growers lost buyer and credit together.

Instead: action 1.

Do not push rust-resistant replanting on credit in an El Niño year

CONTRAINDICATED SKEPTIC; CONFIRMED WITH SOURCING FIXES

Young plants are the most drought-vulnerable, and replanting means three years of debt with no harvest. Starmaya, the hybrid bred in Nicaragua, showed rust in 8% of plants in trials against 1% for Marsellesa; clonal F1 plants cost $0.50–0.60 against $0.20–0.30 for conventional seedlings; seed-grown F1s cup at 80+ against 85+ for clonal (Georget et al. 2026).

Instead: pay for fungicide, fertiliser, and borer traps now; resume replanting after the 2027 rains.

Do not use a yield-only drought trigger for highland Jinotega

CONTRAINDICATED SKEPTIC; VERIFICATION

The highlands had near-normal rain; their El Niño loss is heat, borer, and unfilled beans that show up as a downgrade. A yield trigger would pay nothing while farmers lose money on grading.

* Suggested formulation: check with a health worker before sharing.

Instead: quality triggers in the highlands, rainfall or volume triggers only in the western strip.

Do not send money labelled as a donation or run a branded programme

CONTRAINDICATED ALL LENSES; NATIONAL VERIFICATION

The foreign-agents law reportedly counts funding from foreign companies; the Red Cross was dissolved; clergy have been expelled, including Bishop Rolando Álvarez of Matagalpa. Branded picker top-ups or clean-ups can be reclassified as foreign funding. The Frontline lens: hand anything to a party structure and people stop opening the door.

Instead: price differentials on invoiced coffee, passed through estate payrolls.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

DATE	EVENT	TOLL	WHAT FOLLOWED
15 Oct 1891	Department created, split from Matagalpa; estates from Granada and León families and German and British investors (La Fundadora under the Englishman Charles Potter)	—	Coffee capital from outside sets who absorbs shocks
c. 1960	Lake Apanás flooded; Centroamérica plant	—	Coffee catchment tied to the grid
1980s	Contra war; Contras targeted rural cooperatives and their leaders	Figures for Jinotega not found	Lasting political divisions
Oct 1998	Hurricane Mitch, in La Niña after a record El Niño	At least 3,800 dead nationally; no Jinotega toll found	—
2012–13	Coffee rust; worst on the poorest, least-advised plots	—	Only ~14% of arabica replanted since

* Suggested formulation: check with a health worker before sharing.

DATE	EVENT	TOLL	WHAT FOLLOWED
Nov 2020	Hurricane Iota, in La Niña	3 dead in Wiwilí; Kilambé homes damaged; Bocay, Pantasma, Wiwilí without power and water	Opposition mayors "left alone"
2023	El Niño	Output −10%, exports −15% (USDA), with unfilled beans	—
6 Dec 2023	Mercon bankruptcy; CISA stops buying before harvest	About half of exportable coffee without its buyer	Assets under state administration; lower farm prices
Dec 2025	Picker shortage: El Cuá 40% short; gold mining pays C$1,200 a day	—	Pay raised to C$100+ per lata
Aug–Sep 2026	Dry Corridor deficit; highland rain near normal; government calls it "not severe"	—	This briefing

The recurrence nobody has noticed, from the Historian and confirmed by the rust study: every crisis in Jinotega since 1891 has been survived by whoever held credit, not whoever held knowledge. The lever sits in the export house's pre-harvest ledger, which is also the one channel the state has not closed.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Drought is severe here** versus **the highlands are near-normal**. Both lens sources failed as described. The best evidence splits the department: deficit in the west, quality loss in the highlands.
- **Buyers pay early** versus **exporters may not be solvent**. Resolved together: pay early, through more than one exporter, under letters of credit.
- **Replant now to beat rust** versus **not in an El Niño year**. Resolved toward waiting for the 2027 rains.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED BLIND SPOT

No lens reached the Indigenous Miskito and Mayangna communities near Bosawás, where malaria risk and state neglect are likely highest. No lens counted Jinotega's pickers or found how many latas a day they pick, and the funding rests on both. No lens looked at gold mining as a competing employer beyond its wage, including its mercury and water effects in the same catchments. And the cooperative union the lenses relied on has a website frozen since 2014 and currently down; its real size is unknown.

HOW TO READ THIS METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done

* Suggested formulation: check with a health worker before sharing.

here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

This run. Coffee-industry actions were ranked first at the reader's request. National facts verified in the Ocotal run were given to every lens so they could spend their effort on local evidence. Web search was exhausted; all twelve agents worked by direct fetch. Five verifiers checked thirteen recommendations; the verdict tags carry the result.

The El Niño projection covers *heat only* and publishes nothing for Central America. The Adaptation Inventory holds two Nicaragua entries, both cited, and nothing on coffee, malaria, or harvest labour.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Jinotega, Nicaragua. Climate Impact Lab impact region, as the report records it: Jinotega department, about 480,000–500,000 people (1–2 Climate Impact Lab regions). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak).

* Suggested formulation: check with a health worker before sharing.

Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.

- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (safe water, oral rehydration, and malaria protection in harvest camps) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (verification status in brackets) · Rainfall and livestock, 25 Sep 2026 confidencial.digital · heat, borer, and unfilled beans, 28 Apr 2026 confidencial.digital [partially confirmed; rainfall anecdote struck] · USDA GAIN Coffee Annual NU2025-0004 · NU2024-0005 [confirmed] · Mercon bankruptcy confidencial.digital · CISA aftermath confidencial.digital [confirmed with corrections] · Harvest pause and pickers confidencial.digital · pickers house to house confidencial.digital [corrected] · Villarreyna et al. 2020 doi.org · Starmaya, WCR worldcoffeeresearch.org · Georget et al. 2019 PMC6818232 · Georget et al. 2026 PMC13579250 [confirmed with sourcing fix] · Iota victims confidencial.digital · Iota in Jinotega confidencial.digital · Hurricane Iota wikipedia [corrected] · Jinotega es.wikipedia · Jinotega department es.wikipedia · electricity sector wikipedia [partially confirmed] · Rivera et al. 2017, iCCM PMC6612741 · Mokdad et al. 2015 PMC4327957 · malaria incidence api.worldbank.org · Soto et al. 2022 PMC9142003 · under-5 mortality api.worldbank.org · PAHO El Niño brief 2026 PMC13421129 · Pryce 2018 PMC6418392 · Clasen 2015 PMC4625648 [confirmed] · Mierisch farms fincasmierisch.com · SOPPEXCCA soppexcca.org [stale since 2014] · WFP Nicaragua wfp.org · UNICEF Nicaragua unicef.org · National facts carried from the verified Ocotal briefing (elnino-actions/ocotal-nicaragua-actions.html). Struck or unverified: an August 2026 weather-service rainfall statement (not found); "800–1,000 mm by late April" (anecdote from the south); Jinotega's "65%" or "80%" of national coffee; Apanás "35%" of supply (historical); Peñas Blancas landslide (in Matagalpa, not Jinotega); Contra-war harvest figures; Jinotega malaria data; picker counts; latas per day.
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions; Nicaragua cash transfer and productive grant entry; Nicaragua bridges entry; US contingent loans entry)

* Suggested formulation: check with a health worker before sharing.

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